Shahesta Salama

Computer Science & AI Undergraduate
Data Systems and Trustworthy AI

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EDUCATION

B.Sc. Computer Science & Artificial Intelligence - Alryada University — Sept 2023 - June 2027 (expected)
A grades in Computer Networks, Neural Networks, Data Analysis

RESEARCH PROJECTS

Incident Commander - An SRE Agent That Decides When Not to Act — 2026
1st Place, Student Track, ITIDA DevOps Day 2026 | Archived on Zenodo (DOI) | Paper in preparation | GitHub

- Bayesian belief updating with value-of-information probe selection: **1.2 checks** per incident vs. 2.8 for exhaustive search, at equal accuracy.
- Autonomy depends on reversibility, not confidence: an OPA policy engine outside the agent gates every action; over 63 trials, **10/10** failed fixes auto-reverted and **14/14** irreversible actions were blocked.
- Every decision is sealed in a Merkle tree signed with Ed25519 and verifiable offline. Limits are documented in the repository.

ICVSP - Trust-Aware Cooperative Vehicle Safety Platform — 2026 - 2027
Graduation project, mentored by Valeo engineers | Python, PyTorch, C++ (ESP32-S3), FastAPI | GitHub

- YOLO11n detector fine-tuned on Egyptian day/night footage: **mAP50 0.846** on unseen public images; road-surface and motion filters remove **over 95%** of false speed-bump flags on held-out Egyptian video.
- Trust and consensus engine resisting Sybil and replay attacks cuts false alerts from 46.9% to **0.5%**; full-system simulation with a FastAPI backend: **0% false warnings** across fault and attack scenarios (150+ unit tests).
- Edge-ready: an 8-bit 486 KB model for a low-cost ESP32-S3 unit, with C++ firmware and Ed25519-signed reports.

Fleet Telematics Lakehouse - Ingestion Platform and Event Generator — 2026
Samsung Innovation Campus graduation project (team) | My part: data generator and ingestion platform | GitHub

- FastAPI ingestion API over insert-only PostgreSQL, streamed through Debezium CDC and Kafka into Microsoft Fabric: **3.16M events in 75 min** (700/s, 1.59 s average latency) plus 5.09M batch rows, loaded idempotently.

EXPERIENCE

Samsung Innovation Campus - Big Data & Data Engineering Trainee — Jul - Oct 2026

- Built 15+ data-engineering projects: batch and streaming pipelines with Kafka, NiFi, Spark Structured Streaming, Flink SQL, HDFS and Hive, including a real-time pipeline replacing a 12-hour nightly batch.
- Cloud and operations: AWS (Glue, S3, Athena), Snowflake + dbt, Delta Lake; Airflow and Kubernetes scheduling; Prometheus/Grafana monitoring.

DEPI (IBM Data Science track) - Data Science & Analytics Trainee — Nov 2025 - Jul 2026

- Capstone FinWise AI: inflation forecasting model (Gradient Boosting), SQL database layer and Dash dashboard.

Minly - Software Development & AI Integration Intern — Jul - Aug 2025

- REST APIs and relational schemas in NestJS, TypeORM and PostgreSQL; automated data refresh workflows; Docker.

TECHNICAL SKILLS

Languages: Python, C++, SQL, TypeScript, Bash
AI / ML: PyTorch, Scikit-learn, TensorFlow/Keras, YOLO, SegFormer, OpenCV
Data systems: Spark, Kafka, Flink, Hive, Airflow, dbt, Snowflake, Delta Lake, PostgreSQL
Tools: FastAPI, Docker, Kubernetes, AWS, Linux, Git, GitHub Actions

CERTIFICATIONS & LANGUAGES

Certifications: IBM Data Science Professional (2026) | Supervised ML, DeepLearning.AI (2026) | Huawei/NTI AI Academy (2025)
Languages: Arabic (Native) | English (C1)